

MUHAMMAD OSAMA AMIN

AI Solutions Architect | Cloud Platform Engineering | Applied AI Product Delivery

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PROFILE

AI and cloud architect with six years building production platforms for clients across the Gulf, UK, US and Africa. Founded and scaled an applied AI platform business, shipped commercial GenAI products, and ran 150+ cloud deployments on Azure, AWS and GCP. Deep on LLM orchestration, multi tenant architecture and FinOps, with an 87% infrastructure cost reduction on one production re-platform. Now moving to a full time role.

WORK EXPERIENCE

VM Nebula (Pvt.) Ltd.

Aug. 2024 to Aug. 2026

Founder & Chief Architect | Advisory from Aug. 2026

Islamabad, PK (Remote)

- Ran a cloud and applied AI delivery business serving clients in Saudi Arabia, the UAE, Pakistan and Madagascar; hired and led a team of engineers across infrastructure, DevOps and application delivery.
- Owened DevOps, infrastructure as code and FinOps across AWS, GCP and Azure at an enterprise security and compliance bar, reducing client compute spend by 46%.
- Architected a multi tenant cloud execution platform with tenant isolation, zero trust networking and full observability.
 - Key Result:** re-platformed the production runtime (ECS Fargate to containerised GCE, DynamoDB to Supabase, ECR to Artifact Registry), reducing per unit infrastructure cost by approximately 87%.
- Delivered a voice to ledger operations pipeline that converts spoken Urdu and Pashto into structured database records through Fastify webhooks, Laravel on GCE and Gemini Flash on Vertex AI, giving non technical field staff a zero training data entry channel.

Amin Technologies Limited

Aug. 2022 to Jun. 2025

Cloud Operations Lead | Jan. 2023 to Jun. 2025

Remote (United Kingdom)

- Led cloud operations for a UK registered managed services provider serving enterprise and SME accounts; owned scoping, architecture, implementation and ongoing support.
- Key Results:** built a secure Azure Virtual Desktop estate in the Gulf region (UAE North) with monitoring, private Git and internal chat; delivered multi session FSLogix and RDS profile infrastructure at scale; ran HIPAA compliant cloud hosting for healthcare clients.

Cloud Engineer | Aug. 2022 to Jan. 2023

London, United Kingdom

- Delivered enterprise deployments on Azure, AWS and Google Cloud covering virtual machines, virtual desktop infrastructure and production workloads; built HPC environments improving throughput by 30%; administered Active Directory for 50+ users.

Independent Cloud & Infrastructure Consultant

Nov. 2020 to Present

Freelance / contract delivery

Remote (Global)

- 100+ paid infrastructure engagements for clients across the Gulf, US, UK and Europe at a 4.97 out of 5 rating, with repeat work from more than a dozen clients. Work spans Azure Virtual Desktop estates with VPN gateways and firewall policy, Windows RDS with FSLogix profile containers, AWS and GCP production infrastructure, and backup or disaster recovery design.

Human Development & Peace Organization (HDPO)

Apr. 2023 to Present

Director, IT & Digital (Part time, Remote)

Islamabad, PK

- Own IT systems, tooling, vendor selection and governance; led the digital transformation programme that raised operational efficiency by 35%.

SELECTED PROJECTS & OPEN SOURCE

NebulaSEO: Autonomous AI SEO & Growth Agent

Open source (GPLv3) | 2026

github.com/Vmnebula/NebulaSeo

Python, FastAPI, Next.js 14, GCP

- Built an autonomous Gemini 3 Pro agent with 37 function calling tools that audits Core Web Vitals, ingests Google Search Console and GA4 data through BigQuery, and opens GitHub pull requests to fix meta tags, JSON-LD schema and heading structure with no human in the loop.
- FastAPI agent service plus Next.js 14 TypeScript dashboard, containerised and deployed on Cloud Run with Clerk authentication, Docker Compose and Cloud Build CI.

Substacker: AI Cost Attribution Platform

Live product | Open source (GPLv3) | 2026

github.com/Vmnebula/substacker

substacker.vmnebula.com

- Shipped a live FinOps platform attributing large language model spend to teams and models across OpenAI, Anthropic, Google and Azure. FastAPI backend, Supabase and SQLite persistence, hashed API key auth, WebSocket live updates, budget enforcement, anomaly detection and a published Python SDK.

CORE COMPETENCIES

- Applied AI:** LLM orchestration; retrieval augmented generation; vector databases; structured output and function calling; Vertex AI; AWS Bedrock; Gemini; Claude; voice and speech to structured data pipelines.
- Cloud & Infrastructure:** Azure (AVD/MVD, Entra ID, FSLogix, Windows RDS); AWS (EC2, ECS Fargate, ECR); GCP (GCE, Vertex AI); Active Directory; VPN and firewall policy; zero trust; Docker; infrastructure as code; CI/CD; Linux and Windows Server; FinOps and cost optimisation; HIPAA aligned hosting.
- Engineering:** Python; Node.js; TypeScript and React; PHP (Laravel); PostgreSQL and MySQL; Redis; Supabase; REST APIs.
- Certifications:** Aviatrix Certified Network Engineer; DS4KP Data Science and Machine Learning.

EDUCATION & RECOGNITION

University of Engineering & Technology (UET), Peshawar

2023

B.S., Computer Software Engineering

Peshawar, PK

- Selected for the Misk Accelerator (Saudi Arabia) and Orbit Accelerator; incubated at NIC Peshawar; work featured on national television and BBC Pashto.
- Selected products:** Mizaan AI, a Sharia compliance validation engine for Islamic fintech; Mechanic AI, a self healing DevSecOps agent that patches CVEs and raises pull requests; Scope Shrinker, a voice native AI project manager on the Gemini Live API. github.com/vmnebula